ENSAE Paris - IP Paris

Project

Linear Time Series

ARIMA modeling of a time series

Authors :

Houssem Bouabid
Willy Lin

2024-2025

1 Data

The selected series represents the monthly evolution of cement production from January 1990 to march 2025. This aggregated series is normalized to 100 at the production level in 2021 and adjusted for seasonal variations and working days (CVS-CJO).

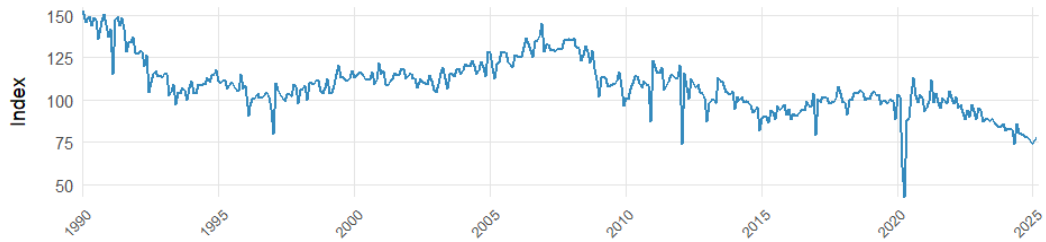


Figure 1: Initial Time Series

The series as shown in the chart seems to be overall decreasing throughout the sample, with minor upswings during some periods of time. In the stretch after 2008, the descent becomes noticeably sharper, reflecting stronger and more sustained downward momentum. We will assess stationarity by first applying the Augmented Dickey–Fuller test, which tests the null hypothesis of a unit root, and then conducting the KPSS test, whose null hypothesis is trend stationarity; a failure to reject the ADF null coupled with a rejection of the KPSS null will provide robust confirmation that the series is non-stationary.

The results of the stationarity tests on the series indicate both that it is non-stationary. The KPSS test, which has the null hypothesis of trend stationarity, rejects the null thus suggesting non-stationarity. Similarly, the Augmented Dickey-Fuller (ADF) test, which tests the null of a unit root (non-stationarity), yields a p-value higher than 0.05, meaning we fail to reject the null. Therefore, the series likely has a unit root and is I(1), but might be stationary after differencing.

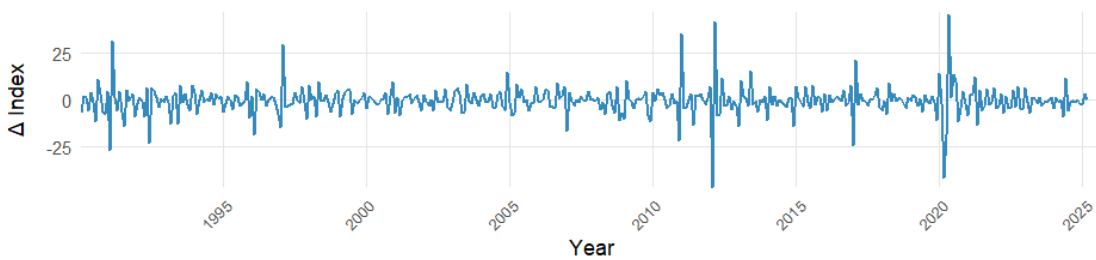


Figure 2: Differenced Time Teries

First-order differencing appears sufficient to render the series stationary, as shown in the second plot of Figure 2. This is further confirmed by the new values of the stationarity tests shown in table A1. This leads us to believe that our series can be represented as follows:

$$X_t = m + c\,t + Y_t$$

where m is the mean and Y_t is a strictly stationary and ergodic process with zero mean. This equation represents our initial series with the initial deterministic trend ct we then applied a first-order difference and subtracted the mean to center the series:

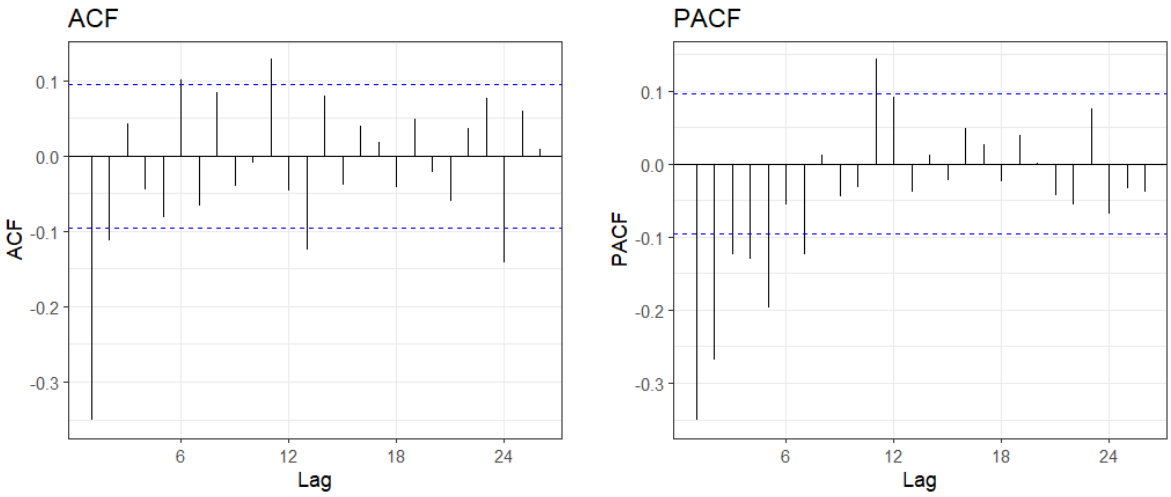
$$\Delta X_t = c + Y_t - Y_{t-1},$$

$$W_t = \Delta X_t - \mathbb{E}[\Delta X_t] = Y_t - Y_{t-1} = \Delta Y_t.$$

Seeing that W_t is a combination of Y_t and Y_{t-1} we conclude that W_t is both strictly stationary and ergodic with zero mean as represented in figure 2.

2 ARMA Modelization

We start by looking for the maximum orders of the AR part and of the MA part using the following autocorrelation and the partial autocorrelation functions plots.



(a) ACF

(b) Partial ACF

Figure 3: Autocorrelation and Partial Autocorrelation

The autocorrelation function (ACF) shown in Figure 3a exhibits a single large negative spike at lag 1 (approximately -0.35), after which all subsequent autocorrelations lie within the 95% confidence bands. Such an abrupt cut-off after the first lag is characteristic of an MA(1) process which indicates a $q_{max} = 1$. On the other side, the partial autocorrelation function (PACF) in Figure 3b displays a slowly decaying sequence of negative coefficients across the first several lags. which suggests that $p_{max} = 7$.

We then proceed to the testing of the different models and assessing their robustness using three main metrics: AICc, AIC and BIC. The table A2 summarizes the results for the different models sorted by AICc. We remark that AICc and AIC have nearly the same values this can be explained by the fact that our number of observations is very large and thus the small-sample correction is negligible. We notice that ARIMA$(7, 1, 0)$ yields the minimum AICc (2811), but its BIC (≈ 2843) is substantially higher than that of ARIMA$(1, 1, 1)$ (AICc $= 2812$, BIC ≈ 2824). That's exactly how BIC is designed: it uses $k \ln n$ rather than $2k$ as the penalty, so it favors the simpler model even though AR(7) "wins" by a hair on raw AICc. A difference of just one point in AICc is negligible, and the lower BIC for $(1, 1)$ clearly favors the much simpler model. Hence, the optimal choice is to retain the ARIMA$(1, 1, 1)$ model.

To determine which model better captures the dependence structure of the series, we applied a Ljung–Box test (up to lag 24) on the residuals of both ARIMA$(1, 1, 1)$ and ARIMA$(7, 1, 0)$. The resulting p-values are presented in Tables A3 and A4. For ARIMA$(1, 1, 1)$, several p-values fall below 0.05 at lags beyond 11, indicating significant autocorrelation of the residuals. In contrast, all p-values for ARIMA$(7, 1, 0)$ exceed 0.05, so we never reject the null of zero autocorrelation through lag 24. Because only ARIMA$(7, 1, 0)$ produces truly uncorrelated residuals at every lag up to 24, it is the preferred model for capturing the series' dependence structure.

Finally, we can test the normality of our residuals in order to determine whether the assumption of Gaussian residuals is plausible. Figure 4 appears to indicate that the residuals follow a distribution close to the standard normal law, since the quantiles of the normal distribution and the quantiles of our residual series are close, except for some extreme outliers.

Coefficient	Estimate	Std. Error	Z value	P value
ar1	-0.5337025	0.04835613	-11.037	0.00×10^{0}
ar2	-0.4363945	0.05458363	-7.995	1.33×10^{-15}
ar3	-0.2957196	0.05703881	-5.185	2.17×10^{-7}
ar4	-0.2751715	0.05723736	-4.808	1.53×10^{-6}
ar5	-0.2680504	0.05686879	-4.713	2.44×10^{-6}
ar6	-0.1155077	0.05436201	-2.125	3.36×10^{-2}
ar7	-0.1207199	0.04818086	-2.506	1.22×10^{-2}

Table 1: Coefficient estimates, standard errors, z-values, and p-values for ARIMA(7,1,0).

All seven AR coefficients are negative and statistically significant (all $p < 0.05$), with the largest absolute values at lag 1 (-0.534) and lag 2 (-0.436). This means each month's change is pulled back by the previous one or two months' changes (and to a lesser but still significant degree by months 3–7), indicating strong short-term negative autocorrelation in the differenced series.

Model representation

Below is the "canonical representation" for the ARIMA$(7, 1, 0)$.

$$\big((1-B)\,X_t - c\big)\,\phi(B) \;=\; \psi(B)\,\varepsilon_t$$

with $\phi(B)$ the AR-polynomial of order 7 and $\psi(B)$ the MA-polynomial. Since we have no MA terms, $\psi(B) = 1$. Concretely:

$$\underbrace{\big((1-B)\,X_t - c\big)}_{\text{differenced series minus drift}} \times \underbrace{\big(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_7 B^7\big)}_{\phi(B)} \;=\; \underbrace{1}_{\psi(B)} \times \varepsilon_t \tag{1}$$

$$\underbrace{\big((1-B)\,X_t - c\big)}_{\text{differenced series minus drift}} \times \underbrace{\big(1 - \mathrm{AR}_1\,B - \mathrm{AR}_2\,B^2 - \cdots - \mathrm{AR}_7\,B^7\big)}_{\Phi(B)} \;=\; \varepsilon_t \tag{2}$$

$$\big(1 - \mathrm{AR}_1\,B - \mathrm{AR}_2\,B^2 - \cdots - \mathrm{AR}_7\,B^7\big)\,(1-B)\,X_t \;-\; c\,\big(1 - \mathrm{AR}_1\,B - \mathrm{AR}_2\,B^2 - \cdots - \mathrm{AR}_7\,B^7\big) \;=\; \varepsilon_t \tag{3}$$

with ε_t being the white-noise and B the backward (lag) operator.

3 Prediction

An ARIMA(7,1,0) with "drift" c can be written in differences as

$$\Delta X_t \;=\; c \;+\; \phi_1\,\Delta X_{t-1} \;+\; \phi_2\,\Delta X_{t-2} \;+\; \cdots \;+\; \phi_7\,\Delta X_{t-7} \;+\; \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2),$$

where we define

$$\Delta X_t \;=\; X_t \;-\; X_{t-1}, \quad t = 8, 9, \ldots, T, \ldots$$

In the appendix B we go through the details of the calculations of the confidence interval. We demonstrate the following expression of the error term vector (true minus forcasted predictions):

$$\mathbf{e} \;=\; \begin{pmatrix} X_{T+1} - \widehat{X}_{T+1|T} \\ X_{T+2} - \widehat{X}_{T+2|T} \end{pmatrix} \;=\; \begin{pmatrix} \varepsilon_{T+1} \\ (1+\phi_1)\,\varepsilon_{T+1} \;+\; \varepsilon_{T+2} \end{pmatrix}.$$

and then by calculating its variance-covariance matrix we obtain the expression of Σ:

$$\boxed{\Sigma \;=\; \sigma_\varepsilon^2 \begin{pmatrix} 1 & (1+\phi_1) \\ (1+\phi_1) & 1 + (1+\phi_1)^2 \end{pmatrix}.}$$

Finally knowing that $det(\Sigma) = \sigma^4_{\epsilon_t} > 0$ we get the following equation for confidence interval as demonstrated in the appendix:

$$\boxed{\left\{ \mathbf{X} \in \mathbb{R}^2 \;\middle|\; (\mathbf{X} - \widehat{\mathbf{X}})^{\mathsf{T}} \, \Sigma^{-1} \, (\mathbf{X} - \widehat{\mathbf{X}}) \;\leq\; \chi^2_{2,\,1-\alpha} \right\}.}$$

where, $\mathbf{X} - \widehat{\mathbf{X}} = \mathbf{e}$ and $\chi^2_{2,\,1-\alpha}$ denotes the $(1-\alpha)$-quantile of χ^2_2

For the set

$$\left\{ X \in \mathbb{R}^2 \;\middle|\; (X - \widehat{X})^T \, \Sigma^{-1} \, (X - \widehat{X}) \;\leq\; \chi^2_{2,\,1-\alpha} \right\}$$

to be a confidence region of level $1-\alpha$, the following conditions must hold:

- The ARIMA model is correctly specified (orders, drift, etc.) and the linear projection on past observations is used to form $\widehat{X}_{T+1|T}$ and $\widehat{X}_{T+2|T}$. And to have a sufficiently large sample size T so that the estimated parameters are consistent.
- The innovations ε_t are exactly i.i.d. $\mathcal{N}(0, \sigma^2_\varepsilon)$.
- The covariance matrix Σ of the forecast errors is invertible.

If any of these assumptions is severely violated (for instance, non-Gaussian errors or a mis-specified ARIMA model), then the assertion

$$(X - \widehat{X})^T \, \Sigma^{-1} \, (X - \widehat{X}) \;\sim\; \chi^2_2 \;\Longrightarrow\; \left\{ (X - \widehat{X})^T \, \Sigma^{-1} \, (X - \widehat{X}) \leq \chi^2_{2,\,1-\alpha} \right\}$$

is no longer rigorously exact, and the nominal coverage $1-\alpha$ may fail to hold.

Confidence interval representation

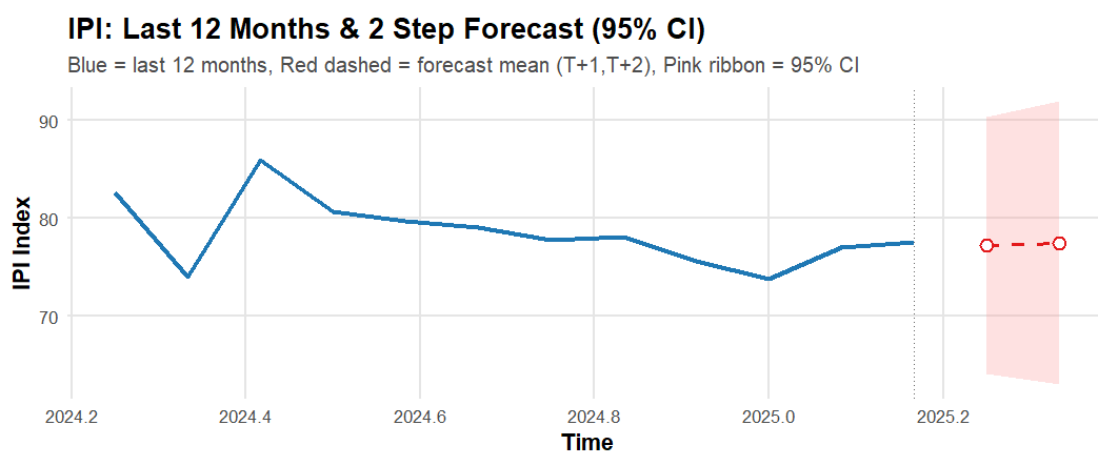


Figure 4: Graphical Representation for the $\alpha = 95\%$ Confidence Interval

When forecasting two steps ahead (T + 2), you face more uncertainty than when forecasting just one step ahead because you must account not only for the shock that will occur at $T+1$ (which has variance σ^2) but also for the separate shock that will occur at $T+2$ (another independent draw with the same variance σ^2). Furthermore, since the $T+1$ prediction is itself based on information up through time T rather than on an observed value, any error at $T+1$ carries over into the $T+2$ forecast. In other words, because the first forecast is "approximately correct" rather than exact, its inevitable error propagates forward, combining with the new shock at $T+2$. As a result, the total forecast error variance for $T+2$ is strictly larger than for $T+1$, which is why the 95% prediction interval widens as we move further into the future. These observations are even clearer when looking at the numerical values summarized in the table 2 where the length of the second interval is $10,34\%$ larger than the first one.

Horizon	Lower95	Upper95	Interval Length
T+1	64.13	90.33	26.20
T+2	63.04	91.95	28.91

Table 2: 95% Prediction Intervals and Interval Lengths for T+1 and T+2

Open Question: Let Y_t be a stationary time series observed for $t = 1, 2, \ldots, T$. We assume that Y_{T+1} is available faster than X_{T+1}. Under which condition(s) does this extra information allow you to improve the prediction of X_{T+1}? How would you test those condition(s)?

We suppose that we have two jointly stationary series, $\{X_t\}$ and $\{Y_t\}$. At time T, we know all past values

$$\mathcal{F}_T \;=\; \{\, X_T,\; X_{T-1},\; \ldots;\; Y_T,\; Y_{T-1},\; \ldots \},$$

and our goal is to predict X_{T+1}. If Y_{T+1} becomes available immediately (before we need our final forecast of X_{T+1}), then our best-possible predictor is

$$\widehat{X}_{T+1} \;=\; \mathbb{E}\big[X_{T+1} \mid \mathcal{F}_T,\; Y_{T+1}\big].$$

Compare that to what we would do without Y_{T+1}:

$$\widetilde{X}_{T+1} \;=\; \mathbb{E}\big[X_{T+1} \mid \mathcal{F}_T\big].$$

Hence, we gain extra information exactly when

$$\mathbb{E}\big[X_{T+1} \mid \mathcal{F}_T,\; Y_{T+1}\big] \;\neq\; \mathbb{E}\big[X_{T+1} \mid \mathcal{F}_T\big].$$

In other words, There is instantaneous Granger causality from $(Y)_t$ to $(X)_t$. The prediction of X_{t+1} greatly improves when we know the actual value of Y_{t+1}.

To test this hypothesis, we can use an instantaneous Granger causality test.

- H_0: No instantaneous causality
- H_1: Instantaneous causality

A Additional Results

Table A1: ADF/KPSS test results

Series	Test	Statistic	p-value
Raw	ADF	-2.1843	0.4996
Raw	KPSS	0.55082	0.01
1st Difference	ADF	-10.204	0.01
1st Difference	KPSS	0.052109	0.1

Table A1 presents the results of the Augmented Dickey–Fuller (ADF) and KPSS trend-stationarity tests, applied first to the raw series and then to its first difference.

p	q	AICc	AIC	BIC
7	0	2811.0	2811.0	2843.0
1	1	2812.0	2812.0	2824.0
5	1	2812.0	2812.0	2841.0
7	1	2813.0	2812.0	2849.0
2	1	2814.0	2814.0	2830.0
5	0	2814.0	2814.0	2838.0
6	1	2814.0	2814.0	2846.0
4	1	2815.0	2815.0	2839.0
6	0	2815.0	2815.0	2843.0
3	1	2816.0	2816.0	2836.0
0	1	2819.0	2819.0	2827.0
4	0	2828.0	2828.0	2848.0
3	0	2833.0	2833.0	2849.0
2	0	2837.0	2837.0	2849.0
1	0	2866.0	2866.0	2874.0
0	0	2919.0	2919.0	2924.0

Table A2: Model selection criteria (AICc, AIC and BIC) for different (p, q) combinations.

lag	pval
1	NA
2	NA
3	0.387945064
4	0.341918557
5	0.222206246
6	0.136572674
7	0.220790271
8	0.074705438
9	0.109467240
10	0.117351648
11	0.016717883
12	0.023176817
13	0.005564850
14	0.007652686
15	0.011821392
16	0.014512488
17	0.020528049
18	0.027864407
19	0.036586384
20	0.045035973
21	0.046028331
22	0.052793383
23	0.057199445
24	0.022908171

Table A3: Ljung–Box p-values (lags 1–24) ARIMA$(1, 1, 1)$

lag	pval
1	NA
2	NA
3	0.7584061
4	0.9483797
5	0.9902855
6	0.9909439
7	0.9912060
8	0.9972129
9	0.9992522
10	0.9797229
11	0.3353587
12	0.4074309
13	0.2046353
14	0.2481753
15	0.3116414
16	0.3254458
17	0.3945561
18	0.4173639
19	0.5198625
20	0.5445960
21	0.5675193
22	0.6171351
23	0.6250942
24	0.3192290

Table A4: Ljung–Box p-values (lags 1–24) for ARIMA$(7, 1, 0)$

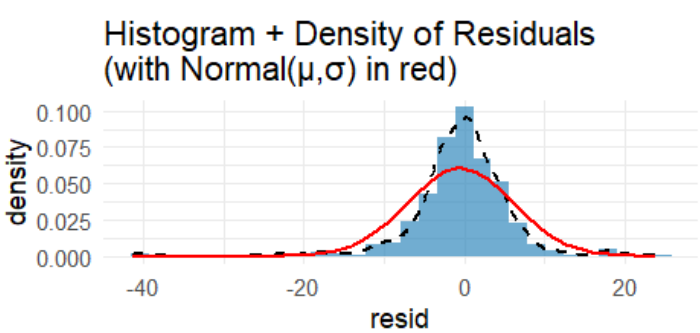


Figure 5: Residuals Distribution

B Confidence Interval Calculation for ARIMA(7,1,0)

One-step-ahead forecast ($h = 1$)

At time T, all of the following are known:

$$X_T,\ X_{T-1},\ \ldots,\ X_{T-7} \implies \Delta X_T,\ \Delta X_{T-1},\ \ldots,\ \Delta X_{T-6}, \quad \text{and } \varepsilon_T.$$

Because

$$\mathbb{E}\big[\varepsilon_{T+1} \mid \mathcal{F}_T\big] \;=\; 0,$$

the optimal (linear) predictor of ΔX_{T+1} is

$$\widehat{\Delta X}_{T+1|T} \;=\; c \;+\; \phi_1\,\Delta X_T \;+\; \phi_2\,\Delta X_{T-1} \;+\; \cdots \;+\; \phi_7\,\Delta X_{T-6}.$$

Hence the one-step-ahead forecast for the level X_{T+1} is

$$\boxed{\widehat{X}_{T+1|T} \;=\; X_T \;+\; \widehat{\Delta X}_{T+1|T} \;=\; X_T \;+\; \big[c \;+\; \phi_1\,\Delta X_T \;+\; \phi_2\,\Delta X_{T-1} \;+\; \cdots \;+\; \phi_7\,\Delta X_{T-6}\big].}$$

the one-step-ahead forecast-error is

$$X_{T+1} - \widehat{X}_{T+1|T} \;=\; \varepsilon_{T+1},$$

Two-steps-ahead forecast ($h = 2$)

Next, we write the model at time $t = T + 2$:

$$\Delta X_{T+2} \;=\; c \;+\; \phi_1\,\Delta X_{T+1} \;+\; \phi_2\,\Delta X_T \;+\; \cdots \;+\; \phi_7\,\Delta X_{T-5} \;+\; \varepsilon_{T+2}.$$

To forecast ΔX_{T+2} from $\mathcal{F}_T$, we substitute ΔX_{T+1} by its forecast $\widehat{\Delta X}_{T+1|T}$. Thus,

$$\widehat{\Delta X}_{T+2|T} \;=\; c \;+\; \phi_1\,\widehat{\Delta X}_{T+1|T} \;+\; \phi_2\,\Delta X_T \;+\; \cdots \;+\; \phi_7\,\Delta X_{T-5}.$$

Accordingly, the level-forecast for X_{T+2} is

$$\widehat{X}_{T+2|T} \;=\; \widehat{X}_{T+1|T} \;+\; \widehat{\Delta X}_{T+2|T}.$$

so the two-step-ahead forecast-error in levels is

$$\begin{aligned} X_{T+2} - \widehat{X}_{T+2|T} \;&=\; \big[X_{T+1} + \Delta X_{T+2}\big] \;-\; \big[\widehat{X}_{T+1|T} + \widehat{\Delta X}_{T+2|T}\big] \\ &=\; \big(X_{T+1} - \widehat{X}_{T+1|T}\big) \;+\; \big(\Delta X_{T+2} - \widehat{\Delta X}_{T+2|T}\big). \end{aligned}$$

$$\begin{aligned} \Delta X_{T+2} - \widehat{\Delta X}_{T+2|T} &= \Big[c \;+\; \phi_1\,\Delta X_{T+1} \;+\; \phi_2\,\Delta X_T \;+\; \cdots \;+\; \phi_7\,\Delta X_{T-5} \;+\; \varepsilon_{T+2}\Big] \\ &\quad - \Big[c \;+\; \phi_1\,\widehat{\Delta X}_{T+1|T} \;+\; \phi_2\,\Delta X_T \;+\; \cdots \;+\; \phi_7\,\Delta X_{T-5}\Big] \\ &=\; \phi_1\,\big(\Delta X_{T+1} - \widehat{\Delta X}_{T+1|T}\big) \;+\; \varepsilon_{T+2} \;=\; \phi_1\,\varepsilon_{T+1} \;+\; \varepsilon_{T+2}. \end{aligned}$$

Hence

$$\boxed{X_{T+2} - \widehat{X}_{T+2|T} \;=\; \varepsilon_{T+1} \;+\; \big(\phi_1\,\varepsilon_{T+1} + \varepsilon_{T+2}\big) \;=\; (1 + \phi_1)\,\varepsilon_{T+1} \;+\; \varepsilon_{T+2}.}$$

Distribution of the two-dimensional forecast-error

Define the 2-vector of true minus forecasted levels:

$$\mathbf{e} \;=\; \begin{pmatrix} X_{T+1} - \widehat{X}_{T+1|T} \\ X_{T+2} - \widehat{X}_{T+2|T} \end{pmatrix} \;=\; \begin{pmatrix} \varepsilon_{T+1} \\ (1+\phi_1)\,\varepsilon_{T+1} \;+\; \varepsilon_{T+2} \end{pmatrix}.$$

Because ε_{T+1} and ε_{T+2} are independent $\mathcal{N}(0, \sigma_\varepsilon^2)$, $\mathbf{e}$ is bivariate normal with mean $\mathbf{0}$ and covariance matrix

$$\Sigma \;=\; \mathrm{Var}(\mathbf{e}) \;=\; \begin{pmatrix} \mathrm{Var}(\varepsilon_{T+1}) & \mathrm{Cov}\big(\varepsilon_{T+1},\, (1+\phi_1)\,\varepsilon_{T+1} + \varepsilon_{T+2}\big) \\ \mathrm{Cov}\big(\varepsilon_{T+1},\, (1+\phi_1)\,\varepsilon_{T+1} + \varepsilon_{T+2}\big) & \mathrm{Var}\big((1+\phi_1)\,\varepsilon_{T+1} + \varepsilon_{T+2}\big) \end{pmatrix}.$$

Therefore,

$$\boxed{\Sigma \;=\; \sigma_\varepsilon^2 \begin{pmatrix} 1 & (1+\phi_1) \\ (1+\phi_1) & 1 + (1+\phi_1)^2 \end{pmatrix}.}$$

Joint $1-\alpha$ confidence region

$$\mathbf{X} \;=\; \begin{pmatrix} X_{T+1} \\ X_{T+2} \end{pmatrix}, \qquad \widehat{\mathbf{X}} \;=\; \begin{pmatrix} \widehat{X}_{T+1|T} \\ \widehat{X}_{T+2|T} \end{pmatrix}.$$

Then $\mathbf{X} - \widehat{\mathbf{X}} = \mathbf{e} \sim \mathcal{N}(\mathbf{0},\, \Sigma)$. Because Σ is positive-definite,

$$(\mathbf{X} - \widehat{\mathbf{X}})^{\mathsf{T}}\, \Sigma^{-1}\, (\mathbf{X} - \widehat{\mathbf{X}})$$

follows a χ^2_2 distribution. Denote by $\chi^2_{2,\,1-\alpha}$ the $(1-\alpha)$-quantile of χ^2_2. Hence a valid joint $1-\alpha$ confidence region for $\mathbf{X}$ is

$$\boxed{\big\{\, X \in \mathbb{R}^2 \;\big|\; (\mathbf{X} - \widehat{\mathbf{X}})^{\mathsf{T}}\, \Sigma^{-1}\, (\mathbf{X} - \widehat{\mathbf{X}}) \;\le\; \chi^2_{2,\,1-\alpha} \big\}.}$$

C Commented Code

Listing 1: Script R pour l'ARIMA de l'IPI (production de ciment)

```r

# Project Title      : <ARIMA Modelization of IPI (cement production)>
# Authors            : <Houssem Bouabid & Willy Lin>
# Date               : <31/05/2025>
# Notes              : At the start of the code (step - 0) we left all the packages
# that we used in comment form if necessary you can uncomment it. All the important
# outputs (test results, plots ...) are in the appendix of pdf report. Finally make
# sure the csv path is correct (step - 2) to correctly run the code.


# 0. necessary packages

# Note: Uncomment the following lines if packages are not already installed
#install.packages(c(
#  "tidyverse",
#  "lubridate",
#  "tseries",
#  "forecast",
#  "seastests",
#  "tidyr",
#  "nortest"
#))

# 1. Load libraries
library(readr)
library(dplyr)
library(ggplot2)
library(lubridate)
library(tseries)
library(forecast)
library(seastests)
library(purrr)
library(tidyr)
library(nortest)


# 2. Import CSV data from INSEE


csv_path  <- file.path("Put your path here")

ipi_raw <- read_delim(
  "Cement_Prod.csv",
  delim      = ";",
  skip       = 4,                  # Skip first 4 lines of metadata
  col_names  = c("Periode","Valeur","Codes"),
  locale     = locale(decimal_mark = ".")
)


# 3. Prepare the date column and numeric values

ipi <- ipi_raw %>%
  select(Periode, Valeur) %>%
  mutate(
    # Convert YYYY MM in Periode to a Date object (first day of month)
    Date       = as.Date(paste0(Periode, "-01")),
```

```r
    # Convert 'Valeur' from character to numeric
    ValeurNum  = as.numeric(Valeur)
  ) %>%
  arrange(Date)


# 4. Determine the last observation date and its value (for subtitle)

last_date  <- max(ipi$Date)
last_value <- ipi$ValeurNum[ipi$Date == last_date]


# 5. Plot the full series with ggplot2 (raw series)

ggplot(ipi, aes(x = Date, y = ValeurNum)) +
  geom_line(size = 0.8, color = "#348ABD") +
  labs(
    title    = NULL,
    subtitle = NULL,
    x        = NULL,
    y        = "Index"
  ) +
  scale_x_date(
    breaks = seq(from = as.Date("1990-01-01"),
                 to   = last_date + years(1),
                 by   = "5 years"),
    date_labels = "%Y",
    expand = expansion(add = c(0, 30))
  ) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.02))) +
  theme_minimal(base_family = "Helvetica") +
  theme(
    panel.grid.major = element_line(color = "grey90"),
    panel.grid.minor = element_blank(),
    # axes
    axis.text.x  = element_text(angle = 45, hjust = 1, size = 8),
    axis.title.y = element_text(margin = margin(r = 10)),
    # fond
    panel.background = element_rect(fill = "white", color = NA)
  )


# 6. Convert the series to a 'ts' object for stationarity tests

ipi_ts <- ts(
  ipi$ValeurNum,
  start     = c(year(min(ipi$Date)), month(min(ipi$Date))),
  frequency = 12
)


# 7. Perform stationarity tests on the raw series

# 7.1 Augmented Dickey-Fuller (ADF) test on the raw series
adf_raw <- adf.test(ipi_ts)
print(adf_raw)
#for this test we obtained a big value => not stationary

# 7.2 (KPSS) test on the raw series
#     null = "Trend" tests for trend stationarity
kpss_raw <- kpss.test(ipi_ts, null = "Trend")
```

```r
print(kpss_raw)
#same with this test the p_value was below the threshold of 0.05 => presence of a trend


# 8. First difference the series

ipi_diff1 <- diff(ipi_ts, differences = 1)  # First difference

# Build a data frame for plotting the differenced series
ipi_diff_df <- data.frame(
  Time      = time(ipi_diff1),
  DiffValue = as.numeric(ipi_diff1)
)

# Plot the first-differenced series
ggplot(ipi_diff_df, aes(x = Time, y = DiffValue)) +
  geom_line(size = 0.8, color = "#348ABD") +
  labs(
    x = "Year",
    y = "delta Index"
  ) +
  scale_x_continuous(
    # Breaks every 5 years from min to max of the differenced time
    breaks = seq(
      floor(min(ipi_diff_df$Time)),
      ceiling(max(ipi_diff_df$Time)),
      by = 5
    ),
    expand = expansion(add = c(0, 0.5))
  ) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.02))
  ) +
  theme_minimal(base_family = "Helvetica") +
  theme(
    panel.grid.major = element_line(color = "grey90"),
    panel.grid.minor = element_blank(),
    axis.text.x  = element_text(angle = 45, hjust = 1, size = 8),
    axis.title.y = element_text(margin = margin(r = 10)),
    panel.background = element_rect(fill = "white", color = NA)
  )


# 9. Perform stationarity tests on the differenced series

# 9.1 ADF test on first difference
adf_diff1 <- adf.test(ipi_diff1)
print(adf_diff1)
#p-value below 0.05 => stationary

# 9.2 KPSS test on first difference (null = "Trend")
kpss_diff1 <- kpss.test(ipi_diff1, null = "Trend")
print(kpss_diff1)  # Note the KPSS statistic and p-value
#p-value above 0.05 => stationary


# 10. Plot ACF and PACF of the differenced series

# 10.1 Autocorrelation Function (ACF)
ggAcf(ipi_diff1) +
  ggtitle("ACF") +
```

```r
  theme_bw()

# 10.2 Partial Autocorrelation Function (PACF)
ggPacf(ipi_diff1) +
  ggtitle("PACF") +
  theme_bw()


# 11. Grid search over ARMA(p,1,q) for p = 0..7, q = 0..1

results <- expand_grid(
  p = 0:7,
  q = 0:1
) %>%
  mutate(
    # Fit ARIMA(p,1,q) on original series (d = 1)
    fit  = map2(p, q, ~ Arima(ipi_ts, order = c(.x, 1, .y))),

    # Extract AIC and BIC from fitted models
    AIC  = map_dbl(fit, ~ AIC(.x)),
    BIC  = map_dbl(fit, ~ BIC(.x)),

    # Compute sample size (n) and number of estimated parameters (k)
    n    = map_dbl(fit, ~ length(residuals(.x))),
    k    = map_dbl(fit, ~ length(coef(.x)) + 1),

    # Compute corrected AIC (AICc)
    AICc = AIC + 2 * k * (k + 1) / (n - k - 1)
  ) %>%
  arrange(AICc) %>%                # Sort by ascending AICc
  select(p, q, AICc, AIC, BIC)     # Keep relevant columns

print(results)  # Display grid search results


# 12. Fit two selected ARIMA models based on grid search

# Fit ARIMA(1,1,1)
fit111 <- Arima(ipi_ts, order = c(1, 1, 1))

# Fit ARIMA(7,1,0)
fit710 <- Arima(ipi_ts, order = c(7, 1, 0))


# 13. Extract residuals from both models

resid_111 <- residuals(fit111)
resid_710 <- residuals(fit710)


# 14. Define function Qtests() to compute Ljung Box p-values up to lag k

Qtests <- function(series, k, fitdf = 0) {
  pvals <- apply(matrix(1:k), 1, FUN = function(l) {
    pval <-
      if (l <= fitdf) {
        # If lag <= number of parameters, return NA
        NA
      } else {
        # Compute Ljung Box test p-value at lag l
        Box.test(
```

```
          series,
          lag    = l,
          type   = "Ljung-Box",
          fitdf  = fitdf
        )$p.value
      }
    return(c("lag" = l, "pval" = pval))
  })
  return(t(pvals))  # Return transpose: each row is a lag and p-value
}


# 15. Compute Ljung Box p-values for lags 1 to 24

LB_table1 <- Qtests(resid_111, k = 24, fitdf = 2)
LB_table2 <- Qtests(resid_710, k = 24, fitdf = 2)

LB_df1 <- as.data.frame(LB_table1)
LB_df2 <- as.data.frame(LB_table2)

LB_df1$lag  <- as.integer(LB_df1$lag)
LB_df1$pval <- as.numeric(LB_df1$pval)
LB_df2$lag  <- as.integer(LB_df2$lag)
LB_df2$pval <- as.numeric(LB_df2$pval)

# Print Ljung box tables
print(LB_df1)
print(LB_df2)


# 16. Time series plot of residuals from ARIMA(7,1,0)

autoplot(resid_710) +
  ggtitle("Residuals from ARIMA(7,1,0)") +
  ylab("Residual") +
  theme_minimal()


# 17. Histogram and density overlay of residuals with Normal distribution

# Compute sample mean and standard deviation of residuals
mu_r    <- mean(resid_710)
sigma_r <- sd(resid_710)

ggplot(data = data.frame(resid = resid_710), aes(x = resid)) +
  # Histogram with density scaling
  geom_histogram(
    aes(y = ..density..),
    bins  = 30,
    fill  = "#348ABD",
    alpha = 0.7
  ) +
  # Empirical density (dashed black line)
  geom_density(
    color    = "black",
    linetype = "dashed",
    size     = 0.8
  ) +
  # Overlay the true normal curve N(mu_r, sigma_r) in red
  stat_function(
    fun   = dnorm,
```

```r
    args  = list(mean = mu_r, sd = sigma_r),
    color = "red",
    size  = 0.8
  ) +
  ggtitle("Histogram + Density of Residuals\n(with Normal distribution(in red)") +
  theme_minimal()


# 18. Extract ARIMA(7,1,0) coefficient estimates, SE, z stats, and p values

# 18.1 Extract point estimates (coefficients)
coefs <- coef(fit710)

# 18.2 Extract standard errors from variance-covariance matrix
ses <- sqrt(diag(fit710$var.coef))

# 18.3 Compute z statistics and two sided p values
zvals <- coefs / ses
pvals <- 2 * (1 - pnorm(abs(zvals)))

# 18.4 Assemble results
results <- data.frame(
  Coefficient = names(coefs),
  Estimate    = coefs,
  StdError    = ses,
  Z.value     = round(zvals, 3),
  P.value     = signif(pvals, 3)
)
print(results)


# 19. Confidence interval question for points T+1 and T+2

# 19.1 Generate forecast object for h = 2 (95% confidence interval)
fc710 <- forecast(fit710, h = 2, level = 95)

print(fc710)

# 19.3 Extract individual components
one_ahead_mean <- fc710$mean[1]
one_ahead_lo95 <- fc710$lower[1, "95%"]
one_ahead_hi95 <- fc710$upper[1, "95%"]

two_ahead_mean <- fc710$mean[2]
two_ahead_lo95 <- fc710$lower[2, "95%"]
two_ahead_hi95 <- fc710$upper[2, "95%"]


# 20. plot

# 20.1 Compute last 12 months data frame
last12_start <- time(ipi_ts)[length(ipi_ts) - 11]  # Numeric time for T-11
last12 <- window(ipi_ts, start = last12_start)

last12_df <- data.frame(
  Time  = time(last12),
  Value = as.numeric(last12)
)

fc_df <- data.frame(
  Time            = time(fc710$mean),
```

```r
  PointForecast  = as.numeric(fc710$mean),
  Lo95           = fc710$lower[, "95%"],
  Hi95           = fc710$upper[, "95%"]
)

# 20.2 Plot last 12 months and forecast intervals
ggplot() +
  # 1) Plot last 12 months as a solid blue line
  geom_line(
    data = last12_df,
    aes(x = Time, y = Value),
    color = "#1f78b4",   # Blue
    size  = 1.2
  ) +

  # 2) Add 95% CI ribbon (pink, alpha = 0.3) for the 2 step forecast
  geom_ribbon(
    data = fc_df,
    aes(x = Time, ymin = Lo95, ymax = Hi95),
    fill  = "#fb9a99",   # Light pink
    alpha = 0.3
  ) +

  # 3) Add dashed red line for forecast mean
  geom_line(
    data     = fc_df,
    aes(x = Time, y = PointForecast),
    color    = "#e31a1c",  # Red
    linetype = "dashed",
    size     = 0.9
  ) +

  # 4) Add circle markers at the two forecast points
  geom_point(
    data   = fc_df,
    aes(x = Time, y = PointForecast),
    color  = "#e31a1c",
    fill   = "white",
    shape  = 21,
    size   = 3,
    stroke = 1
  ) +

  # 5) Add vertical dotted line at time T to separate observed and forecast
  geom_vline(
    xintercept = max(time(ipi_ts)),
    linetype   = "dotted",
    color      = "gray40",
    size       = 0.5
  ) +

  # 6) Titles and theme changes
  labs(
    title    = "IPI: Last 12 Months & 2 Step Forecast (95% CI)",
    subtitle = paste0(
      "Blue = last 12 months, ",
      "Red dashed = forecast mean (T+1, T+2), ",
      "Pink ribbon = 95% CI"
    ),
    x = "Time",
    y = "IPI Index"
```

```r
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12, color = "gray30"),
    axis.title    = element_text(face = "bold"),
    panel.grid.major = element_line(color = "gray90"),
    panel.grid.minor = element_blank()
  )


# 21. Compute lengths of forecast intervals


# 21.1 Calculate interval length (Hi95 - Lo95) and add as a new column
fc_df$IntervalLength <- fc_df$Hi95 - fc_df$Lo95

# 21.2 (Optional) Reorder columns for clarity: Time, PointForecast, Lo95, Hi95,
    IntervalLength
fc_df <- fc_df[, c("Time", "PointForecast", "Lo95", "Hi95", "IntervalLength")]

# 21.3 Print the resulting data frame
print(fc_df)
```